

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY****Seventh Semester of B. Tech (IT) Examination****May 2015****IT405 Advanced Computing****Date: 01.05.2015, Friday****Time: 1:30 p.m. To 4:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

**SECTION – I****Q - 1 Answer the below given multiple choice questions: [07]**

- a) Which of the following service is used for setting password less communication between nodes in a cluster.  
(a) NFS (b) MPICH (c) SSH (d) All of the above
- b) Hosting an application on Google App Engine is an example of  
(a) Software as a service (b) Platform as a service  
(c) Infrastructure as a service (d) Both (a) and (b)
- c) What is used to logically assign and separate physical resources such as memory and CPU in a cloud computing model?  
(a) Bios (b) Hypervisor (c) Load balancer (d) Blade chassis
- d) In cloud computing, which term is used to describe the nature of IT infrastructure and allows dynamic scaling depending on the demand for resources?  
(a) Elasticity (b) Clustering (c) Consolidation (d) Virtualization
- e) SETI@home(Search for Extraterrestrial Intelligence)project is an example of  
(a) Cluster Computing (b) Grid Computing  
(c) Cloud Computing (d) Hybrid Computing
- f) Which is not a major cloud computing platform?  
(a) Google 101 (b) IBM Deep blue (c) Microsoft Azure (d) Amazon EC2
- g) Clusters are homogeneous whereas Grids are \_\_\_\_\_.  
(a) Heterogeneous (b) Homogeneous (c) Distributed (d) None of the above

**Q – 2.a** There are 5 nodes in a cluster with response time of 10,20,30,40 and 50 nano seconds respectively. To expand the cluster 5 more nodes with response time of 20 nano seconds each are to be added. Will the performance of new system improve or degrade as a whole? **[04]**

**Q – 2.b** Explain the role of RMS (Resource Management and Scheduling) technique in cluster computing. [07]

**OR**

**Q – 2.b** Why system monitoring is required in cluster? Which factors should be monitored? [07]

**Q – 2.c** What is Cluster Tuning? [03]

**OR**

**Q – 2.c** PVM can function as Resource Manager. Justify the statement. [03]

**Q - 3 Write Short Note On:(Any two)** [14]

- a. Flexible Load Sharing Algorithm.
- b. Multi Site Data Grid
- c. Virtual Organization

## **SECTION – II**

**Q - 4 Do as directed:** [07]

- a. Define Isoefficiency. [01]
- b. Give the difference between cluster, grid and cloud [03]
- c. Why Grid is consider as next generation internet? How Grid provides restricted access to resources? [03]

**Q – 5.a** Write the essential characteristics of Cloud computing. [04]

**Q – 5.b** Explain the service model of cloud computing. [05]

**OR**

**Q – 5.b** What is Amazon EC2? List the services provided by it. [05]

**Q – 5.c** How is Cross site trust management handled in grid computing? [05]

**OR**

**Q – 5.c** Which are the major grid topologies? Elaborate computational grid and data grid in detail. [05]

**Q – 6 Attempt any Two:** [14]

- a. Describe the deployment models of cloud computing.
- b. What is Hadoop? Explain the architecture of Hadoop.
- c. What is Google App engine?. Give the steps to develop the application using Google App engine.

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